This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Submitted by **AI Applied Consortium** March 15, 2025
www.aaiconsortium.org

INTRODUCTION

The AI Applied Consortium (Consortium)represents the unified voice and collective expertise of influential organizations spanning diverse sectors, including leading technology firms, prominent academic institutions, innovative startups, Fortune 500 corporations, and engaged public-sector entities. We are committed to fostering innovation, facilitating strategic public-private partnerships, and accelerating practical, real-world applications of (AI). In response to the Request for Information on developing the U.S. AI Action Plan we submit this document to assert that expertise, articulate a shared vision, and demonstrate leadership in AI.

By effectively translating AI technologies into tangible solutions across key sectors, supportive policy frameworks will enable sustained investment, innovation, and industry growth, reinforcing and prioritizing the United States' global leadership in AI. Such strategic policy alignment is essential to ensuring the U.S. remains at the forefront of technological advancements, attracts world-class talent and investment, and establishes international standards for responsible, ethical, and impactful AI development.

Our unified and proactive posture allows us to effectively advocate for balanced policy frameworks, pathways to sustainable innovation, robust workforce development, and enhanced national security through strategic public-private collaborations and targeted investments.

VOICE, EXPERTISE AND INFLUENCE OF THE AI APLIED CONSORTIUM

The Consortium brings together the deep expertise, strategic networks, and broad market reach of over 30 distinguished Trustees and Advisors, representing 25 globally impactful organizations across 11 key sectors: Technology, Financial Services, Manufacturing, Academia, Oil & Gas, Petrochemicals, Wholesale Distribution, Entertainment, Engineering Services, Consulting, and the Public Sector. This leadership is bolstered by contributions from graduate and PhD students at our academic partner institutions, infusing cutting-edge research and fresh perspectives into our efforts.

Together, our coalition represents a market influence spanning organizations whose combined economic activity nears $1 trillion USD annually. Through Fortune 500 companies, world-class universities, influential public-sector entities, and leading technology innovators, the Consortium is uniquely positioned to shape evidence-based policy, accelerate sector-specific innovation, and set international benchmarks for secure, ethical, and impactful AI deployment

A 360° VIEW OF AI: A MULTI-DIMENSIONAL APPROACH TO POLICY AND LEADERSHIP

To effectively shape national AI policy, the United States must adopt a 360° view of AI. A comprehensive, multi-dimensional approach that integrates the perspectives of four foundational pillars:

1. Academic Institutions and Research Leaders, who bring critical insights into emerging technologies, ethics, and long-term societal impact.
2. Enterprise and Industry Practitioners, the organizations deploying AI in real-world settings across supply chains, healthcare, energy, finance, and manufacturing, who understand practical use cases, adoption barriers, and operational risks.
3. Technology Developers and Solution Providers, who create the foundational models, platforms, infrastructure, and tools that drive AI innovation and scalability.
4. Public Sector Leaders and Policy Architects, including regulatory agencies and government bodies responsible for crafting frameworks that ensure safety, equity, innovation, and national security.

An effective AI Action Plan must draw from all four of these interconnected domains, ensuring policies are grounded in academic rigor, informed by real-world applications, technically feasible, and aligned with national strategic priorities.

PRIORITY AREAS AND RECOMMENDED ACTIONS

The AI Applied Consortium's recommendations span six core policy areas, shaped by a 360° perspective that brings together academic leaders, industry experts, technologists, and public sector voices. This approach ensures each recommendation is grounded, actionable, and nationally relevant.

The six focus areas are:

1. **AI Infrastructure and Hardware**
2. **AI Model Development and Open Foundations**
3. **Security, Cybersecurity, and Data Privacy**
4. **Education and Workforce Development**
5. **Energy Efficiency and Sustainability**
6. **Innovation, Intellectual Property, and Competitiveness**

These priorities reflect cross-sector input and aim to advance U.S. leadership in AI through responsible growth, resilience, and global competitiveness.

1. AI INFRASTRUCTURE AND HARDWARE

ACADEMIC RESEARCH LEADERS

The academic community plays a foundational role in translating national AI ambitions into scientific and technological breakthroughs. Institutions require access to high-performance hybrid compute infrastructure, particularly modular systems that integrate CPU and GPU environments to support cutting-edge research in Physics-Informed Machine Learning (PINNs), Graph Neural Networks (GNNs), and Neural Operators. These approaches enable domain-informed model development that is both interpretable and data-efficient

Recommended Policy Action: Provide federal funding and shared access programs to equip universities with modular HPC infrastructure tailored to interdisciplinary research.

Rationale: Science ML approaches reduce training data burdens and improve model generalization and interpretability, thus accelerating breakthroughs in domains like climate science, materials, and biology.

Evidence & Case Studies: The Department of Energy's *Summit* supercomputer, through the COVID-19 HPC Consortium, enabled rapid simulation of molecular interactions to identify potential therapeutics—demonstrating how hybrid HPC systems accelerate discovery using domain-informed AI. Similar approaches are now advancing climate modeling, materials science, and genomics. Source – U.S. Department of Energy

Potential Benefits: Accelerated discovery cycles, increased participation in national AI objectives, and improved return on research investment.

Alignment & Collaboration: Supports the National AI Initiative Act and NAIRR roadmap. Fosters collaboration between academic institutions, NSF AI Institutes, and federal agencies.

INDUSTRY PRACTITIONERS

Industry deployment of AI hinges on reliable, secure, and scalable infrastructure across supply chains, logistics, and manufacturing. Despite the potential, adoption is constrained by infrastructure literacy gaps, fragmented procurement pathways, and high barriers to ownership

Recommended Policy Action: Implement tax incentives, federal grants, and regulatory modernization to incentivize AI infrastructure ownership and integration across industrial sectors.

Rationale: Infrastructure readiness, including clean data pipelines, real-time processing, and workforce preparedness, is a prerequisite for operational AI deployment.

Evidence & Case Studies: According to the U.S. Census Bureau and SBA, 98–99% of U.S. manufacturers are SMEs (SBA). While 51.6% now report having an AI strategy, 74% still struggle to scale AI value across operations (Deloitte, 2024), (BCG, 2024). Pilot programs with infrastructure-linked training show higher AI deployment rates (USCAM, 2024).

Potential Benefits: Increased productivity, reduced bottlenecks, enhanced workforce resilience, and national competitiveness.

Alignment & Collaboration: Aligns with CHIPS and Science Act goals and Manufacturing USA strategy. Builds frameworks for public-private collaboration and regional equity.

TECHNOLOGY DEVELOPERS

Developers require modular, performant environments that enable experimentation, model training, and real-time deployment. Infrastructure must support parallelization, cloud-native compute orchestration, and GPU/CPU scheduling. These technical capabilities are critical to model robustness and deployment at scale

Recommended Policy Action: Fund the co-development of open-source infrastructure frameworks and standards for compute orchestration and modularity.

Rationale: Integrated platforms improve interoperability and transparency while supporting traceable, reproducible experimentation.

Evidence & Case Studies: Open orchestration frameworks enhance GPU/CPU efficiency and workflow performance. The open-source **Taskflow** framework improved execution speed by 29%, memory usage by 1.5×, and throughput by 1.9× on standard multicore systems (arXiv). NSF AI Institutes similarly demonstrate how modular, orchestrated environments accelerate research velocity

Potential Benefits: Accelerated innovation cycles, enhanced transparency, and open innovation through reusable frameworks.

Alignment & Collaboration: Supports CHIPS and Science Act and federal open-source AI initiatives.

PUBLIC SECTOR & POLICY ARCHITECTS

Strategic public investment in AI infrastructure is vital to national resilience, economic growth, and security. Government must enable this through aligned legislation, multi-stakeholder governance, and modernized procurement.

Recommended Policy Action: Align infrastructure policy with the CHIPS and Science Act and National AI Initiative Act by advancing public-private partnerships and streamlining regulatory pathways.

Rationale: Shared models like Manufacturing USA show how pooled resources and federal backing drive scalable innovation.

Evidence & Case Studies: Global firms have invested over **$15B in AI startups**, citing infrastructure as a key enabler (WEF, 2024). The DOE's **Summit supercomputer** accelerated COVID-19 vaccine modeling (U.S. DOE, 2021). The **NAIRR Pilot** demonstrates the potential of shared infrastructure to broaden access to compute and data (nairrpilot.org).

Potential Benefits: Boosted competitiveness, faster innovation, and stronger national security through domestic AI infrastructure.

Alignment & Collaboration: Supports the NAIRR Pilot, AI Executive Orders, and the CHIPS and Science Act by coordinating infrastructure across federal, academic, and industry ecosystems.

2.AI MODEL DEVELOPMENT AND OPEN SOURCE

ACADEMIC RESEARCH LEADERS

AI model development within academic settings increasingly relies on science-based machine learning approaches that embed physical laws, experimental data, and cross-disciplinary insights. Techniques like PINNs (Physics-Informed Neural Networks), GNNs (Graph Neural Networks), and Neural Operators require training environments that support explainability, uncertainty quantification, and heterogeneous data integration

Recommended Policy Action: Develop a national AI testing and validation framework, including regulatory sandboxes for academia to test models under varying levels of scrutiny.
Rationale: These structured environments allow institutions to develop robust models while ensuring interpretability, fairness, and safe deployment, particularly in high-impact domains such as climate, materials, and healthcare.
Evidence & Case Studies: The DARPA XAI program demonstrates the federal interest in explainability. DOE science ML programs are advancing drug discovery and materials science by improving model efficiency and reducing data burdens (DOE Office of Science).
Potential Benefits: Faster innovation cycles, improved public trust, and integration of ethical AI from inception.
Alignment & Collaboration: Supports ongoing NSF AI Research Institutes initiatives, the NAIRR Pilot, and open-source reproducibility mandates that promote academic-industry collaboration and rigorous scientific AI development.

INDUSTRY PRACTITIONERS

Enterprises face challenges in safely operationalizing AI models across sectors such as finance, energy, and healthcare due to the complexity of managing high-risk applications and regulatory ambiguity
Recommended Policy Action: Create tiered regulatory frameworks based on application risk, combined with incentives for participation in AI sandboxes and post-deployment reporting.
Rationale: Risk-tiered policy allows businesses to deploy low-risk AI rapidly while maintaining oversight on sensitive domains like healthcare and autonomous vehicles.
Evidence & Case Studies: The EU AI Act and NIST AI Risk Management Framework both support risk-based governance. The U.K.'s pro-innovation framework enables regulators to apply common AI principles within sector-specific contexts, accelerating adoption while balancing compliance and innovation.
Potential Benefits: Greater clarity for implementation, protection from reputational or legal risks, and expanded market trust.
Alignment & Collaboration: Aligns with the Executive Order on AI , which calls for removing regulatory barriers to innovation and developing a national AI action plan. Complements international frameworks by emphasizing risk-informed, scalable governance and public-private regulatory collaboration.

TECHNOLOGY DEVELOPERS

Developers need freedom to build and test open-source AI models without fear of violating shifting compliance boundaries. Yet open-source models also require clear accountability, provenance tracking, and abuse mitigation, especially for generative AI.
Recommended Policy Action: Establish federal guidance on responsible open-source development, including provenance metadata standards, red-teaming requirements, and usage reporting.
Rationale: Transparent, traceable development pipelines ensure safety while maintaining the open innovation that has driven much of AI's growth.

Evidence & Case Studies: The Safety by Design for Generative AI outlines safety-by-design in generative AI. Research by Thorn emphasizes real risks in misuse without appropriate red-teaming and content separation protocols.
Potential Benefits: Safer deployment of generative models, reduced risk of harm, and stronger global interoperability.
Alignment & Collaboration: Supports collaboration across IEEE and open-source foundations like LF AI.

PUBLIC SECTOR & POLICY ARCHITECTS

Government plays a pivotal role in ensuring AI models, especially those deployed in critical infrastructure, public services, or defense, are safe, traceable, and legally compliant
Recommended Policy Action: Implement national explainability and transparency standards for AI in high-impact sectors, while supporting sandbox-based evaluation pathways for broader adoption.
Rationale: Models used in law enforcement, healthcare, and finance must be auditable, fair, and explainable to retain public trust.
Evidence & Case Studies: DARPA's XAI program shows how explainability improves operational reliability. IEEE's P7002: Data Privacy Process standard provides guidance for documenting data provenance and supporting transparency in automated decision systems.
Potential Benefits: Public trust, minimized bias and discrimination, stronger regulatory clarity.
Alignment & Collaboration: Reinforces federal priorities under FTC AI Guidance, and the Executive Order on AI

3. SECURITY, CYBERSECURITY, AND DATA PRIVACY

ACADEMIC RESEARCH LEADERS

Academic researchers emphasize the need for rigorous standards in AI model integrity, especially as AI tools become foundational to sectors like healthcare, education, and national infrastructure. Security-by-design must be embedded into AI development pipelines from the earliest research stages. This includes adversarial testing protocols, privacy-preserving training techniques (e.g., federated learning), and formal methods for verifying model robustness. Universities and federally funded research centers should serve as testbeds for scalable cybersecurity frameworks and model auditability studies
Recommended Policy Action: Fund security-first academic research programs in AI safety and model integrity, with direct incentives for publishing reproducible protocols.
Rationale: Academic institutions can pioneer standardized testing methodologies, establish open benchmarks, and drive international best practices when properly resourced.
Evidence & Case Studies: Programs like DARPA's GARD show the role of academia in developing adversarial resilience frameworks.
Potential Benefits: Improved national resilience, early threat detection, and a pipeline of AI talent trained in safety-by-design.

Alignment & Collaboration: Supports the NSF's **Secure and Trustworthy Cyberspace (SaTC) program** and aligns with research priorities outlined in the CHIPS and Science Act of 2022 (H.R.4346),

INDUSTRY PRACTITIONERS

As enterprises scale AI adoption, they face growing legal, reputational, and financial risks from deepfakes, algorithmic fraud, and AI-enabled malware. Governance structures must evolve to include model audit boards, ethics oversight, and real-time incident response. Leading organizations are also investing in data lineage, system-level logging, and continuous model testing.

Recommended Policy Action: Incentivize enterprise-wide AI governance frameworks through regulatory alignment and grant-backed compliance support.

Rationale: Federal guidance and financial incentives can help businesses adopt responsible AI without stifling innovation in high-value sectors like finance, retail, and logistics.

Evidence & Case Studies: The FTC's AI Compliance Plan outlines transparency and accountability practices for federal AI use. Real-world incidents like deepfake-enabled fraud illustrate the consequences of weak enterprise safeguards.

Potential Benefits: Reduced legal exposure, improved consumer trust, and a more resilient cyber posture.

Alignment & Collaboration: Supports CISA's AI Roadmap, aligns with the 2025 Executive Order on AI Innovation, and encourages cybersecurity-aligned workforce development.

TECHNOLOGY DEVELOPERS

Model developers, especially in open-source and generative AI domains, are at the center of growing concerns over IP misuse, algorithmic abuse, and content-based harm. Red-teaming, watermarking, and safety-by-design strategies must be systematically embedded into development pipelines Developers also need clearer federal guidance on content provenance, malicious model use, and hosting obligations.

Recommended Policy Action: Implement federal standards for AI red-teaming, content moderation APIs, and traceable open-source model registries.

Rationale: Developers face mounting pressure to self-regulate with limited resources or consensus. Government frameworks can provide clarity while supporting responsible innovation.

Evidence & Case Studies: Initiatives like OpenAI's Preparedness Framework reflect industry commitment to proactive mitigation.

Potential Benefits: Safer open-source ecosystems, minimized platform liability, and global alignment on AI safety practices.

Alignment & Collaboration: Aligns with NIST AI RMF and IEEE working groups.

PUBLIC SECTOR & POLICY ARCHITECTS

Federal agencies must lead by example, establishing clear AI cybersecurity policies for critical infrastructure, public safety, and digital services. A unified national AI security framework should

include standards for safe deployment, supply chain assurance, and cross-agency threat intelligence sharing.
Recommended Policy Action: Establish a federal AI Security & Privacy Directorate to oversee standards, incident response, and interagency collaboration.
Rationale: A centralized authority ensures coherence across the sprawling landscape of federal AI applications, including defense, transportation, and public health.
Evidence & Case Studies: CISA's AI security roadmap demonstrates how cross-agency coordination and secure-by-design practices can reduce systemic AI risk.
Potential Benefits: Reinforced national cyber resilience, improved public sector trust, and stronger digital infrastructure integrity.
Alignment & Collaboration: Supports international norms such as the OECD AI Principles, aligns with CISA's AI roadmap, and complements interagency models like the NIST AI Risk Management Framework.

4. EDUCATION AND WORKFORCE DEVELOPMENT

ACADEMIC RESEARCH LEADERS

Building national AI fluency begins in the classroom and must extend across every level of education. Academic leaders emphasize embedding AI literacy into K–12 curricula, expanding postsecondary programs, and integrating cross-disciplinary AI education into the arts, humanities, engineering, and sciences alike. This requires federal support to train educators, modernize teaching materials, and make AI-related resources accessible across income brackets and geographies
Recommended Policy Action: Expand federal funding for AI educator training, interdisciplinary curriculum development, and inclusive education resource distribution from K–12 to postsecondary institutions.
Rationale: Without foundational AI literacy, the U.S. risks falling behind global competitors in workforce readiness and technological innovation. Early exposure to AI principles can democratize opportunity and broaden participation in future-facing sectors.
Evidence & Case Studies: The World Economic Forum Future of Jobs Report 2025 highlights that 50% of all employees will need reskilling by 2025 and that 70% of employers plan to hire staff with new skills. The report also notes that AI and automation are expected to transform 39% of current skill sets by 2030.
Potential Benefits: Improved workforce mobility, broader innovation participation, and enhanced talent pipelines for national R&D.
Alignment & Collaboration: Reinforces the National AI Initiative Act and aligns with NSF AI Research Institutes.

INDUSTRY PRACTITIONERS

The U.S. manufacturing sector, particularly small and medium-sized enterprises (SMEs), lags in AI adoption due to a significant skills gap. Workforce training must be inclusive of frontline workers, managers, and executives alike, offering modular programs tailored to real-world use cases such as predictive maintenance, supply chain analytics, and robotics integration

Recommended Policy Action: Provide matching grants and incentives to industry-led workforce development programs in partnership with academic institutions and unions, with a focus on underserved SME ecosystems.
Rationale: AI transformation requires workforce participation across all levels. Without inclusive training, digital adoption stalls, productivity gains are limited, and competitive positioning erodes.
Evidence & Case Studies: USCAM's 2024 Manufacturing Report highlights how workforce-linked AI training significantly increases adoption rates, particularly among SMEs not yet ready to invest in large-scale infrastructure.
Potential Benefits: Increased productivity, higher technology adoption rates, and equitable participation in the digital economy.
Alignment & Collaboration: Supports the NSF National AI Research Institutes, which prioritize education and workforce development as a core pillar. Aligns with industrial reskilling goals under the CHIPS and Science Act, and reinforces national efforts to grow AI talent pipelines through partnerships with academia, labor, and regional innovation ecosystems.

TECHNOLOGY DEVELOPERS

AI-enabled systems across smart cities, autonomous vehicles, and digital infrastructure are driving demand for new skillsets across operations, data governance, and AI system maintenance. Developers require a workforce trained in AI system lifecycle management—including model retraining, deployment safety, and continuous monitoring.
Recommended Policy Action: Expand federal investment in modular, credential-based AI training pathways that allow continuous learning and upskilling aligned with evolving technology platforms.
Rationale: Technology shifts outpace traditional degree programs. Credentialing systems must offer flexibility, speed, and relevance to emerging AI job functions.
Evidence & Case Studies: Reports from CompTIA indicate that non-degree certifications now account for 30–40% of tech workforce credentials, with growing demand in AI-adjacent fields such as data science, infrastructure automation, and cybersecurity. Workforce learners increasingly favor flexible, short-form credentials aligned with evolving AI technology platforms (CompTIA, 2023 Workforce and Learning Trends Report).
Potential Benefits: Higher job readiness, lower retraining costs, and faster workforce adaptation to AI ecosystems.
Alignment & Collaboration: Reinforces public-private collaboration through NSF Industry–University Cooperative Research Centers (IUCRCs), which support scalable workforce development and technology transfer in emerging AI fields.

PUBLIC SECTOR & POLICY ARCHITECTS

The federal government must lead a comprehensive national AI workforce strategy that enables economic inclusion, public sector transformation, and national security readiness. This includes

modernizing federal employment roles, reforming civil service AI pay bands, and establishing job mobility frameworks for public AI professionals.
Recommended Policy Action: Institutionalize an AI workforce roadmap that spans federal agencies, state governments, public schools, and national labs, supported by sustained investments and workforce equity initiatives.
Rationale: AI cannot be siloed to tech firms. The public sector must become an engine for equitable upskilling, regulatory capacity-building, and talent mobilization across underserved communities.
Evidence & Case Studies: According to the WEF Future of Jobs Report, 85% of employers plan to invest in workforce upskilling by 2025, but public-sector roles remain less competitive due to compensation and training lags.
Potential Benefits: Increased talent retention, stronger public capacity for AI regulation and deployment, and equitable economic growth.
Alignment & Collaboration: Reinforces the goals outlined in the National AI Research Resource (NAIRR), aligns with CHIPS and Science Act provisions for workforce preparedness, and complements OSTP AI Workforce Development priorities. Supports global AI education efforts through partnerships like the AI for Good initiative.

5. ENERGY EFFICIENCY AND SUSTAINABILITY

ACADEMIC RESEARCH LEADERS

Energy efficiency and sustainability in AI deployment benefit from foundational research in simulation-driven methods that optimize material use, emissions forecasting, and climate impact modeling. Physics-informed ML approaches, such as PINNs and Neural Operators, enable more accurate predictions in energy systems and infrastructure maintenance, reducing reliance on resource-intensive trial-and-error methods. Research programs that integrate domain knowledge with ML models can accelerate breakthroughs in sectors like climate modeling, grid optimization, and materials science
Recommended Policy Action: Expand funding for science-based ML research focused on sustainability modeling and simulation, and tie research grants to measurable environmental impact indicators.
Rationale: These models reduce energy consumption in both R&D and industrial processes and enable real-time monitoring for emission reductions.
Evidence & Case Studies: DOE's applied science programs have demonstrated that machine learning-accelerated simulations can significantly reduce experimentation cycles, improving prototyping efficiency and helping cut emissions and waste in sectors such as energy and materials science (DOE Office of Science).
Potential Benefits: Lower carbon footprint in industrial AI adoption, improved research translation into climate-positive outcomes, and reduced infrastructure energy overhead.
Alignment & Collaboration: Supports the National AI Research Resource (NAIRR), which promote cross-sector collaboration in AI research and environmental impact. Also reinforces

global efforts under the AI for Earth initiative to advance AI applications in environmental conservation.

INDUSTRY PRACTITIONERS

Energy-intensive sectors such as refining and chemicals can enhance sustainability and operational safety by adopting AI-driven control systems and transitioning from prescriptive compliance to predictive, performance-based frameworks. Predictive models reduce emissions and improve material efficiency while minimizing downtime and hazards. Voluntary AI-powered reporting frameworks modeled after successful aviation data-sharing systems have already shown value in risk forecasting and emissions tracking

Recommended Policy Action: Develop incentives for voluntary reporting frameworks powered by AI and support pilot programs in high-emission industries.

Rationale: Replacing manual, compliance-heavy processes with predictive, AI-driven models empowers companies to act on real-time sustainability insights.

Evidence & Case Studies: The aviation sector's ASIAS program led to significant accident reductions and operational savings. Similar frameworks can be adapted for energy and chemical sectors using AI-based data collection and analysis (FAA ASIAS).

Potential Benefits: Improved environmental stewardship, cost savings, enhanced safety, and competitive advantage through operational excellence.

Alignment & Collaboration: Supports EPA Smart Sectors, DOE sustainability efforts, and interagency infrastructure modernization goals.

TECHNOLOGY DEVELOPERS

In urban and industrial contexts, AI is enhancing energy efficiency through smart city platforms, real-time traffic management, dynamic load balancing, and predictive maintenance. AI-driven control layers reduce energy waste and extend asset lifecycles across transportation, power grids, and water infrastructure. Generative design models and reinforcement learning further optimize energy usage in complex systems.

Recommended Policy Action: Expand funding for edge-AI and embedded sustainability models in infrastructure systems, including grid, water, and mobility sectors.

Rationale: AI can autonomously regulate consumption and predict resource demand, ensuring resilience in critical systems and lowering emissions.

Evidence & Case Studies: AI-driven smart city solutions are improving energy efficiency and reducing resource use. McKinsey reports up to **20%** reductions in commuting times and enhanced urban sustainability. The IEA highlights AI's role in accelerating energy innovation and optimizing systems, advancing technologies like battery materials and carbon capture (McKinsey Report, IEA Report).

Potential Benefits: Lowered emissions, improved resource utilization, and extended lifespan of infrastructure assets.

Alignment & Collaboration: Aligns with DOE Smart Grid efforts to optimize energy distribution and grid management using AI and NSF Smart and Connected Communities to improve sustainability through AI applications in infrastructure.

PUBLIC SECTOR & POLICY ARCHITECTS

AI policy must actively transition environmental regulation from reactive, manual compliance models to AI-enabled, performance-based frameworks. These systems reduce regulatory overhead while enabling more dynamic, real-time enforcement of emissions standards and resource conservation goals. Federal agencies must provide standards, oversight, and coordination to ensure equitable implementation and technological accountability.
Recommended Policy Action: Create an AI-for-sustainability legislative toolkit that includes emissions tracking standards, incentives for predictive AI reporting, and digital compliance frameworks.
Rationale: Compliance models based on real-time AI analytics will be more efficient, reduce fraud, and allow regulatory agencies to focus on high-risk cases.
Evidence & Case Studies: The UK's transition to risk-based safety regulation in heavy industry improved both responsiveness and operational outcomes — U.S. agencies can adopt similar AI-enabled frameworks with federal support (UK HSE case study).
Potential Benefits: Reduced emissions, increased regulatory agility, and enhanced national resilience.
Alignment & Collaboration: Aligns with CHIPS and Science Act.

6. INNOVATION, INTELLECTUAL PROPERTY, AND COMPETITIVENESS

ACADEMIC RESEARCH LEADERS

Academic institutions remain central to frontier innovation, but breakthroughs in AI model design, simulation science, and automated discovery must be accompanied by streamlined technology transfer processes and protected intellectual capital. While high-performance infrastructure is foundational, so too are legal and operational frameworks that promote innovation while safeguarding attribution and ownership.
Recommended Policy Action: Support the commercialization of federally funded AI research through innovation testbeds and IP acceleration frameworks embedded in university ecosystems.
Rationale: Many academic breakthroughs fail to reach the market due to slow tech transfer and lack of IP clarity. Encouraging early-stage proof-of-concept development and shared IP protocols can close this innovation gap.
Evidence & Case Studies: According to the Association of University Technology Managers (AUTM), universities that implement structured tech transfer offices and IP support programs report a threefold increase in licensing agreements and significantly higher startup formation—especially in AI-intensive and data-driven research domains (AUTM, 2023 Annual Licensing Survey).
Potential Benefits: Faster deployment of federally funded AI research, increased university-industry collaboration, and strengthened U.S. leadership in global research commercialization.

Alignment & Collaboration: Aligns with the Bayh-Dole Act. Supports public-private research partnerships and tech transition pathways.

INDUSTRY PRACTITIONERS

Enterprises driving AI innovation need regulatory clarity on IP attribution, particularly in generative and collaborative model development contexts. Moreover, sustained competitiveness depends on the ability to safely commercialize AI products while protecting proprietary algorithms and datasets.

Recommended Policy Action: Develop standardized national frameworks for AI IP attribution, co-developed with industry and legal experts, and embed IP assurance clauses into public-sector R&D funding mechanisms.

Rationale: As AI systems generate novel content and interact with third-party data, questions of authorship, liability, and reuse rights are growing. Clearer IP norms can accelerate innovation by reducing legal ambiguity.

Evidence & Case Studies: Legal uncertainty in AI-generated content has stalled product releases and prompted caution in open collaboration. The EU AI Act includes early examples of model ownership guidance and content provenance.

Potential Benefits: Greater legal certainty, higher rates of AI commercialization, and reduced friction in innovation partnerships.

Alignment & Collaboration: Aligns with U.S. IP law modernization initiatives, and AI R&D priorities under the CHIPS and Science Act.

TECHNOLOGY DEVELOPERS

For AI developers, especially in open-source ecosystems, the lack of consistent attribution, IP licensing clarity, and enforcement mechanisms can stifle innovation. Developers also face rising concerns around IP theft and unauthorized model replication across borders.

Recommended Policy Action: Mandate baseline content provenance tools and digital signature mechanisms in federally supported AI toolkits and open-source registries.

Rationale: Responsible innovation requires visibility into where models and their outputs originate. Watermarking, licensing metadata, and usage audit trails offer scalable safeguards.

Evidence & Case Studies: Open-source platforms embedding licensing metadata (e.g., Hugging Face Model Cards) have shown improved reusability and legal clarity.

Potential Benefits: Greater innovation confidence, improved enforcement of ethical use, and enhanced global trust in U.S. open-source leadership.

Alignment & Collaboration:
Supports NIST AI Risk Management Framework and global interoperability initiatives.

PUBLIC SECTOR & POLICY ARCHITECTS

To sustain U.S. AI leadership, national policy must promote innovation ecosystems while updating IP regimes and competitive safeguards. This includes addressing strategic IP risks, such as exfiltration of AI models, model inversion, and knowledge extraction from deployed systems

Recommended Policy Action: Integrate AI competitiveness metrics and IP protection standards into federal innovation policy, while supporting cross-agency coordination to track global IP risks.
Rationale: Innovation flourishes in environments with clear rules and robust protections. Competitive advantage can be lost if novel models or datasets are compromised before commercial maturity.
Evidence & Case Studies: The DeepSeek.ai case, where leaked LLM weights undermined proprietary model investment, is a cautionary example of unprotected IP eroding billions in R&D value.
Potential Benefits: Greater resilience in AI innovation, higher domestic IP retention, and stronger alignment with national economic priorities.
Alignment & Collaboration: Aligns with the National AI Strategy, NIST Cybersecurity Framework, and international IP treaty modernization efforts.

ADDITIONAL CONSIDERATIONS

INNOVATION VELOCITY

AI is evolving at an unprecedented rate, surpassing the adoption curve of many innovative technologies before it. Its disruptive capacity spans nearly every sector, challenging traditional regulatory, workforce, and innovation processes. Policy frameworks must account for this velocity by supporting adaptive, real-time governance and flexible deployment pathways that enable safe experimentation without stifling growth.

Federal action should recognize that the speed of AI innovation is not just a competitive factor but a strategic imperative. Responsive policy mechanisms, accelerated approval pipelines, and agile public-private collaboration will be essential to ensure the U.S. remains a global leader as AI continues to reshape scientific discovery, national security, and economic competitiveness.

AI ENABLED ECONOMIC RESILIENCE

AI is rapidly becoming foundational to economic resilience, enabling real-time response to market volatility, labor disruptions, and global supply chain instability. Its ability to model complex systems, forecast shocks, and optimize operations positions AI as a critical buffer against systemic risks across sectors.

Federal action should treat AI not just as a catalyst for growth but as core infrastructure for economic continuity. Policy must incentivize AI integration into essential systems—manufacturing, energy, logistics, while ensuring access for SMEs and regional economies. Adaptive funding models, resilience-focused AI testbeds, and cross-sector scenario planning will be key to hardening U.S. economic systems against future disruptions.

CONCLUSION

The AI Applied Consortium affirms that the United States is at a pivotal juncture in shaping AI's trajectory to advance national prosperity, security, and global leadership. As AI rapidly transforms

every facet of the U.S. economy, from infrastructure and innovation to education and governance, it is also revolutionizing long-established sectors like manufacturing, logistics, and energy.
AI's dual-use nature is central to this shift. The same technologies driving commercial innovation also underpin critical defense capabilities, making AI a strategic asset that demands proactive, forward-looking policy.

To lead responsibly, the U.S. must adopt frameworks that are adaptive, innovation-driven, and risk-calibrated, reflecting AI's speed, dual-use dynamics, and potential for both economic growth and systemic risk. Coordinated public-private action is essential to build the infrastructure, regulatory environment, and workforce needed for sustainable leadership.
The Consortium calls for federal action to support modular infrastructure, scalable workforce development, open but accountable AI systems, and strong protections for security, IP, and civil rights. AI must be recognized as vital to economic resilience, enabling agile responses to supply chain disruptions, climate risks, and geopolitical challenges.

As a trusted cross-sector body, the AI Applied Consortium stands ready to support federal partners with strategic guidance, implementation expertise, and cross-disciplinary coordination to translate policy into actionable, lasting impact.

Contributors:

Trustees and Advisors of the Consortium

CONTACT FOR ADDITIONAL INFORMATION:

Konrad Konarski
Chairperson
AI Applied Consortium
[redacted]